

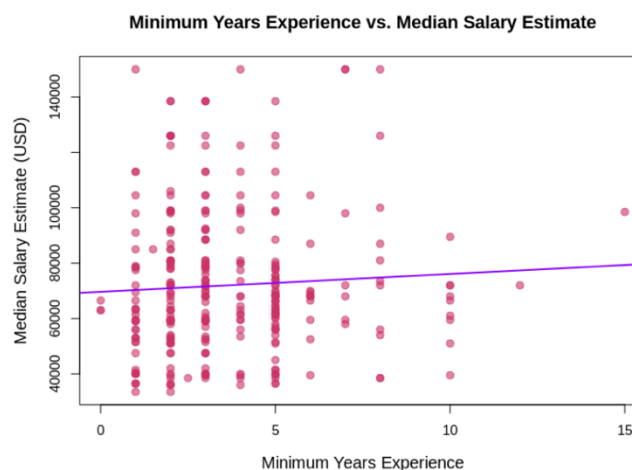
Analyzing Job Market Data with LLMs in Sheets and AI Reporting Agents

Amina Ben Nejma

06.01.2026

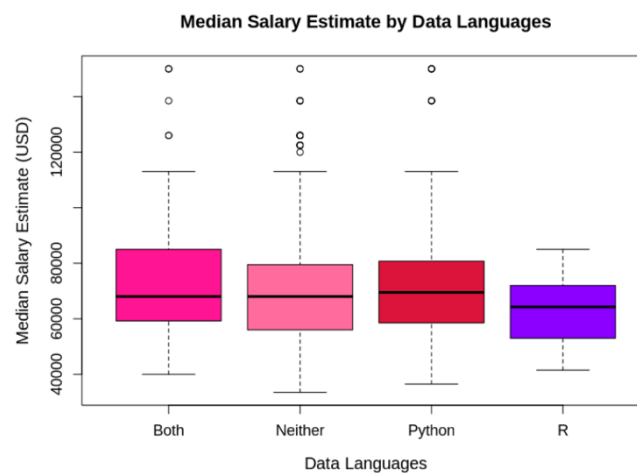
This analysis examines job postings to explore how experience requirements and technical skills relate to salary expectations in the current job market. The project aims to investigating the relationship between years of experience and salary, as well as how experience in data programming languages (R and Python) correlates with earning potential.

Minimum years of experience and salary



This plot shows a positive relationship between minimum years of experience and salary. As the minimum years of experience increases, the median salary tends to increase as well. The violet trend line shows this slight upward trajectory. However, most positions cluster between 0-5 years of experience with salaries ranging from \$40,000-\$100,000. Due to limited data, interpreting the trend beyond 5 years of experience is more difficult and would become more accurate if the data set would be expanded

Salary distributions across different data language requirements



This plot compares salary distributions across different data language requirements (Python and R). Positions requiring both R and Python (dark pink) and Python-only skills (red) show the highest median salaries, while R-only positions (violet) have the lowest median. The “Neither” category (light pink) shows the greatest variation as it includes many lower-paying roles, as well as higher-paying positions. As a result, this category also shows a relatively high overall median salary due to the wide range of salaries represented.

This analysis concludes that while being fluent in both Python and R can be beneficial, when it comes to salary, experience in Python alone may be sufficient. These results have to be interpreted with caution, as the sample size was limited and if being extended could vary from these results.

Reflection

This week’s work made me realise why many of my project, that I tried working out with AI didn’t work. I was aware that there are limitations to the AI but couldn’t figure out what the problem is. While I now know, how important it is to send in a specific and accurate prompt, it is also important to send in a neat document which the AI has to interpret and work on. Other than that, there is still the limitation of the results, which you get from the AI being true. As we’ve seen in the pre-work if we need the AI to search something in the web it might not give us the right answer and if you aren’t

aware of the true answer you wouldn't know. Concluding this means, while it's really helpful to use to clean up your data it's still not to be trusted blindly.